

ARIEL LUBONJA

PhD Candidate, Scalable ML & HPC. Graduation Dec, 2027

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EXECUTIVE BRIEF

I optimize Machine Learning for throughput and latency. I'm experienced with CPU and GPU code profiling, CUDA, GPU acceleration, and Machine Learning Systems.

RESEARCH EXPERIENCE

Runtime-aware Histogramming for Oblique Forests

📅 05/2025 - Present

I reduced the runtime of Oblique Random Forests by 80%—from 12h to 2.5h—enabling our Biomedical team to analyze bigger genomics datasets.

Oblique Forests surpass Random Forests' expressivity, while maintaining interpretability, but are expensive to train.

- Improved by 65% the runtime of [Google's C++ Oblique Forest Library](#) (YDF) by using Binomial projection sampling. [Incorporated into the new version of YDF](#).
- Optimized a further 30% by adaptively discretising data based on workload, with <0.3% accuracy difference. Submitted to KDD 2026.
- Supervised a Masters and 1st year PhD student.

NVIDIA PhysicsNemo

📅 11/2025 - 02/2026

I implemented new, 60% faster functionality in [NVIDIA PhysicsNemo](#). PhysicsNemo is a framework for Physics ML, approximating expensive dynamical systems. It utilizes the [Johns Hopkins Turbulence Database](#) (JHTDB), the world leader in Turbulence simulation datasets. I am a funded member of JHTDB.

- Implemented & verified the new API in NVIDIA PhysicsNemo.
- Achieved 60% runtime reduction by fixing redundant call for low-resolution data.
- Merged in 2 Pull Requests (1, 2).

Scalable Node Embedding for Graph ML

📅 12/2023 - 05/2024

I reduced the billion-edge node embedding runtime from 30 minutes to 56 seconds. Graph ML is a recently popular method that depends on quality Node Embeddings. Existing random walk-based embeddings (e.g. node2vec) cost $O(nk)$ time - prohibitively expensive for large graphs.

- Reduced the billion-edge graph runtime from 30 minutes to 56 seconds by parallelizing across graph edges.
- Our [Graph Encoder Embedding](#) provides a single-pass, parallelizable method.
- Published in [IPDPS 2024](#).

GPU-Accelerated Orthogonal Matching Pursuit

📅 05/2022 - 01/2023

I implement a novel, up to 83x faster GPU acceleration of Scikit-learn Orthogonal Matching Pursuit (OMP). Scikit-learn offers accessible but slow sparse reconstruction, single-threaded processing, making it unsuitable to all but the smallest problems.

- Applied Gramian matrix precomputation and Cholesky factorization to improve single-core runtime by up to 83x.
- Implemented novel GPU acceleration for large matrices by harnessing cuBLAS.
- [Published to arXiv](#). [Github](#) starred by field researchers.

EDUCATION



PhD, Computer Science

Johns Hopkins

📅 08/2022 - 12/2027

- PhD Advisors: Randal Burns, Joshua Vogelstein, Alan Yuille

GPA

3.92 / 4.0



Masters, Data Science

Johns Hopkins

📅 01/2021 - 08/2022

GPA

3.8 / 4.0

SKILLS

High Performance Computing

C++	Python	CUDA	Zarr	Dask
MPI	PyTorch	Git	SLURM	
GPU Profiling		Nsight Systems		

Machine Learning & Mathematics

Tree Ensembles	Graph Learning
Sparse ML	Numerical Optimization

WORK EXPERIENCE

Software Developer

jBoxers

📅 05/2017 - 01/2021 📍 Sofia, Bulgaria

🌐 jboxers.com

- Built an [Alice in Wonderland Chatbot](#) for the [University of Southern California](#) using Speech-to-Text open-source tools, custom NLP model and Google Chatbot
- Developed frontend (Angular) and backend (Django, JavaEE) solutions for the [Library of Congress's Virtual Card Catalog](#).
- Supervised 2 summer interns

ACHIEVEMENTS



Vice President of the Upsilon Pi Epsilon Chapter at JHU



Fellowship to Johns Hopkins University

Fulbright-style prestigious full-ride grant by my home country. 1 of 14 awardees.